PQC Migration (2) — Crypto Agility as Governance; the NIST Path; the Decisions

IKP session 1-7 · Workshop W2 of the PQC Migration Prize series · with NIST participation

Blockchain Governance Initiative Network

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BGIN Block #15 | Georgetown University Capitol Campus, Washington, D.C. | 15 October 2026, Room B, 11:30–13:00 | hybrid

The decisions session: seat the evaluation committee, take the scheme-scope rulings, move the metric register from v0 to v1 · facts follow bgin-global/pqc-agility-competition, draft v0 · ecosystem-neutral: no chain and no scheme is endorsed

This Session — What It Is For

Second IKP working session on the PQC Migration Prize — the decisions session. Crypto agility as a governance property and the NIST PQC signature migration path across Bitcoin, Ethereum and other ecosystems. Block 15 is the second workshop (W2) of the six-month series on the pathway from competition to BGIN Standard to ISO/TC 307.

1. Seating the evaluation committee (Phase A): nominations, conflict-of-interest disclosures, chairs
2. Scheme-scope rulings for the testbed: ML-DSA, SLH-DSA, FN-DSA, XMSS, LMS and the classical baselines — recorded as decision records
3. Metrics register v0 → v1: consolidating the feedback from session 1-4
4. Reference CPU class and reference transaction mix / block — anchors for the performance metrics
5. Whether consensus-layer and user-level evidence are scored on one rubric or two
6. Proposed open-benchmark model for calibration tracks (decision record 0001)
7. Pathway and next steps: testbed workshop (Nov), rubric, “PQC-ready chain” definition and call-text input package at the SSR co-located workshop (13 Dec, Baltimore), BGIN Standard outline (Feb 2027) toward ISO/TC 307

🔗 **Takeaway:** Leave with a seated committee, a v1 metric register, scheme rulings on record, and named owners for the open questions — not a locked migration spec.

Session 1-7 — Agility Is Harder Than Readiness

Evidence on the **mechanism of switching** — hybrid and staged migration, scheme substitution cost, rollback, policy flexibility — because the next migration is already certain and the primitive list will move again.

Metric	What	Unit	How measured
M-14	Agility switch cost	time + coordination steps	scenario: swap scheme A to B under the proposed mechanism — the core agility metric
M-13	Rollback cost	time + coordination steps	scenario: revert a staged migration
M-12	Migration throughput	keys / accounts per unit time	scenario; documents the user vs node-operator burden split
M-21	Legacy verification continuity	pass / fail + node cost	pre-migration signatures still verify after cutover
M-20	Governance fit	structured narrative	transition design, dormant assets, the owner-versus-attacker race — recorded, not ruled

🔗 **Takeaway:** Agility written as a governance property is what lets the same rubric travel: competition → BGIN Standard → ISO/TC 307 → Japan's crypto-asset security criteria.

The NIST Signature Migration Path — What the Register Leans On

NIST is the algorithm and parameter authority for everything in scope; it is not a competition lane. Measurement-methodology cooperation is sought where appropriate, and NIST participation is invited in this session.

- **FIPS 204 ML-DSA**, **FIPS 205 SLH-DSA** — standardized; scheme rows S-01 and S-02.
- **FIPS 206 FN-DSA (Falcon)** — draft; S-03 is confirmed at the scope ruling.
- **SP 800-208** XMSS and LMS / HSS — stateful hash-based; S-04 and S-05 with state handling scored. Non-SP-800-208 parameterizations (hash-based aggregation with ZK-friendly hashes) need an explicit ruling.
- **ACVP** / **CAVP** wording is the correctness gate for calibration tracks.
- Hybrid / composite signatures follow the **IETF LAMPS** direction (C-01); the on-ramp signature candidates are **out of scope by gate G1**.

🔗 **Takeaway:** The register tracks NIST's list; it never extends it. When the list moves, the agility metrics are what say whether a stack can follow.

What Block 15 Decides — the W2 Outputs

Decision	Method	Lands as
Committee seated (Phase A); chairs elected; second convener named	nominations + COI forms; formal vote	COMMITTEE.md roster
OUTCOMES O1–O6 ratified (if not at W1)	formal vote, quorum of half	decisions/ record
Metric register v0 → v1; directions per metric	rough consensus on the PR	evaluation/METRICS.md
Scheme-scope rulings: C-01 hybrid, C-02 quantum-resistant output type as reference, S-04 non-SP-800-208 parameters, C-03 / C-04 / C-05	formal vote per ruling	decisions/ records
Open-benchmark model for calibration tracks	vote on decision 0001; recusals recorded	decisions/0001 accepted
Open questions 1–6 assigned owners	conveners call it	workshop W2 record

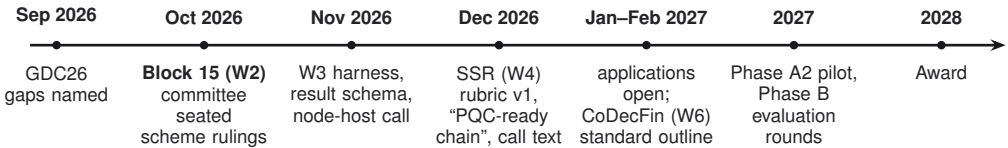
🔗 **Takeaway:** Discuss on Discourse, record by pull request, rule by decision record. Every vote and every scope ruling gets an ADR. Nothing is final until the committee ratifies it and the host publishes the call.

The Open Questions That Need Owners Today

1. Is the headline quantity **total overhead** or **migration speed**? The rubric decides the weights.
2. Who must upgrade first — **node operators or users** — and how does M-12 reflect the burden split?
3. How to score the **owner-versus-attacker race** during a migration window (M-20)?
4. Which **reference CPU class** and **reference tx mix** / **block** anchor M-03 to M-09? Decision record 0002 territory, with *k* and the tolerance band.
5. Are **consensus-layer** (ETH-cons) and **user-level** (BTC-tx, ETH-user) evidence scored on one rubric or two?
6. For each metric, which attack class — **on-spend, at-rest, on-setup** — does it defend against?

Also for the room: what the *"PQC-ready chain" definition* has to contain for M-22 to be scorable by December; and whether the ECDSA.fail public-note discipline (model and harness named, re-run by a validator) is the admission rule for agent-produced numbers.

From This Room to the 2028 Award



Competition / evaluation → BGIN Standard → ISO/TC 307 → Japan’s crypto-asset security criteria

Dates are indicative until the host confirms the call. SSR 2026 PQC-for-blockchains workshop: Sun 13 Dec, Baltimore; peer-reviewed papers there are intended to be reflected in the competition specification.

🔗 **Takeaway:** This morning can still change what the call asks teams to prove. SSR on 13 December is the next drafting stop.

How to Engage

- **Evaluation committee** — nominate via the issue template; seats are individuals with COI disclosure, not projects.
- **Second convener** — the record cannot move on one signing key; a second convener with a key restores the approval rule.
- **Testbed node host** — a university or neutral lab; the call opens at W3 (November); hosting is separated from judging.
- **SSR workshop, 13 Dec** — rubric v1, the “PQC-ready chain” definition, and the call-text input package.
- **Submit** / **build** — when the official call opens (planned 2027).

This session
bgin-global.org/events/20261015-block15/sessions/1-7

Repository
github.com/bgin-global/pqc-agility-competition

Discourse category
bgin.discourse.group/c/post-quantum-crypto-agility-competition/44

🔗 **Takeaway:** Discuss on Discourse, record by pull request, rule by decision record. Every vote and every scope ruling gets an ADR.

Agility is harder than readiness, and it is what future-proofs the system.

The register tracks NIST's list; it never extends it. When the list moves, the agility metrics are what say whether a stack can follow.

BGIN IKP Working Group · Post-Quantum Crypto Agility Competition category on the BGIN Discourse · github.com/bgin-global/pqc-agility-competition

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